

ITBAYAT, Philippines

WMO: 981320

Lat: 20.787N

Lon: 121.840E

Elev: 124

StdP: 99.84

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	18.4	13.2	9.6	19.4	14.5	10.4	19.5	8.8	19.5	8.2	20.9	5.6	40	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	4.1	31.7	28.5	31.2	28.3	30.8	28.0	29.0	31.2	28.7	30.8	28.3	30.4	2.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.3	25.0	30.8	28.0	24.5	30.6	27.7	24.0	30.1	95.3	31.3	93.7	30.9	92.1	30.6	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	7.7	6.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			14.4	29.7	2.3	0.5	12.7	30.0	11.4	30.3	10.1	30.6	8.4	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.6	21.9	22.5	23.5	25.6	27.3	28.4	28.3	27.9	27.4	26.4	25.0	22.8	(6)	
		DBStd	2.74	1.72	1.90	2.05	1.82	1.42	1.19	1.08	1.08	0.95	1.15	1.37	1.84		
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
		HDD18.3	1	1	1	0	0	0	0	0	0	0	0	0	0		
		CDD10.0	5690	370	350	418	468	536	553	567	554	522	507	449	396		
		CDD18.3	2650	112	118	160	218	278	303	308	296	272	249	199	138		
		CDH23.3	22977	168	287	675	1684	2868	3591	3656	3359	2897	2184	1223	385		
		CDH26.7	5856	3	10	61	297	790	1313	1287	1021	709	292	67	6		
(14)	Wind	WSAvg	3.5	4.3	3.9	3.7	3.2	2.6	2.6	3.1	3.1	3.2	4.0	4.1	4.5	(14)	
(15)	Precipitation	PrecAvg	2429	156	87	59	57	213	207	259	338	333	296	251	152	(15)	
(16)		PrecMax	3372	369	268	129	226	616	488	495	607	912	629	808	393	(16)	
(17)		PrecMin	1570	27	17	12	4	7	11	34	80	48	22	49	32	(17)	
(18)		PrecStd	431	92	62	30	46	143	122	139	150	186	162	172	85	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	26.8	27.8	29.6	30.8	32.0	32.6	32.1	31.6	30.9	30.1	28.9	27.6	(19)		
(20)		0.4% MCWB	23.8	25.1	26.0	27.2	28.5	29.0	28.8	28.6	28.1	27.3	26.6	24.1	(20)		
(21)		2% DB	25.6	26.7	28.2	29.9	31.2	31.8	31.5	31.0	30.2	29.2	28.1	26.5	(21)		
(22)		2% MCWB	23.3	24.2	25.4	27.0	28.2	28.6	28.4	28.3	27.7	26.9	26.0	24.4	(22)		
(23)		5% DB	24.9	25.8	27.1	29.2	30.5	31.2	31.0	30.5	29.7	28.7	27.5	25.8	(23)		
(24)		5% MCWB	22.8	23.6	24.8	26.5	27.9	28.4	28.1	28.0	27.3	26.5	25.6	23.9	(24)		
(25)		10% DB	24.2	25.0	26.2	28.4	29.9	30.7	30.5	30.0	29.2	28.2	27.0	25.2	(25)		
(26)		10% MCWB	22.3	23.1	24.1	26.0	27.5	28.0	27.8	27.7	27.0	26.1	25.1	23.4	(26)		
(27)		0.4% WB	24.5	25.5	26.5	27.9	29.1	29.6	29.2	29.1	28.7	27.7	27.0	25.6	(27)		
(28)		0.4% MCDB	26.1	27.3	28.9	29.9	31.2	31.9	31.4	30.9	30.4	29.4	28.5	26.6	(28)		
(29)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	23.7	24.7	25.7	27.2	28.5	29.1	28.8	28.6	28.1	27.2	26.3	24.9	(29)		
(30)		2% MCDB	25.2	26.3	27.7	29.4	30.7	31.3	31.1	30.5	29.9	28.8	27.7	26.0	(30)		
(31)		5% WB	23.1	23.9	25.0	26.7	28.1	28.7	28.4	28.2	27.6	26.8	25.9	24.3	(31)		
(32)		5% MCDB	24.5	25.5	26.8	28.8	30.3	30.9	30.6	30.1	29.4	28.4	27.2	25.5	(32)		
(33)		10% WB	22.6	23.2	24.4	26.1	27.6	28.2	28.0	27.9	27.2	26.4	25.4	23.8	(33)		
(34)		10% MCDB	24.0	24.7	26.1	28.0	29.7	30.4	30.2	29.7	28.9	27.9	26.7	25.0	(34)		
(35)	Mean Daily Temperature Range	MDBR	2.9	3.4	3.6	4.0	4.2	4.1	4.0	3.8	3.5	2.9	2.6	2.7	(35)		
(36)		5% DB MCDBR	3.8	4.6	4.9	5.0	5.3	4.8	4.6	4.5	4.1	3.5	3.3	3.5	(36)		
(37)		5% DB MCWBR	2.8	3.4	3.0	2.9	3.3	2.9	2.9	2.8	2.7	2.4	2.3	2.4	(37)		
(38)		5% WB MCDBR	3.3	4.3	4.5	4.7	5.1	4.5	4.4	4.2	3.9	3.4	3.1	2.9	(38)		
(39)		5% WB MCWBR	2.9	3.5	3.1	3.0	3.4	3.0	3.0	2.8	2.8	2.6	2.4	2.4	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.451	0.453	0.537	0.523	0.455	0.449	0.450	0.470	0.483	0.482	0.480	0.487	(40)		
(41)		taud	2.243	2.267	2.023	2.113	2.392	2.422	2.411	2.338	2.271	2.227	2.215	2.132	(41)		
(42)		Ebn at Noon	816	842	783	799	847	845	844	832	818	803	781	762	(42)		
(43)		Edh at Noon	129	133	176	163	122	117	119	128	135	137	132	140	(43)		
(44)	All-Sky Solar Radiation	RadAvg	3.29	4.21	4.82	5.78	5.93	6.21	6.00	5.41	5.12	4.63	3.62	2.92	(44)		
(45)		RadStd	0.34	0.55	0.49	0.57	0.49	0.53	0.51	0.49	0.54	0.29	0.38	0.44	(45)		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page